

Statistical Data Analysis

Lecture 5: The Bootstrap – part 1

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Periods 4 & 5

◀◀ Chapter 4: Density estimation

📍 Chapter 5: The bootstrap

- › Section 5.1 – Simulation
- › Section 5.2 – Bootstrap estimators for a distribution
- ▶▶ Section 5.3 – Bootstrap confidence intervals
- ▶▶ Section 5.4 – Bootstrap tests
- ▶▶ Section 5.5 – Limitations of the bootstrap

Simulation

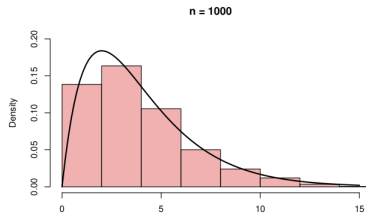
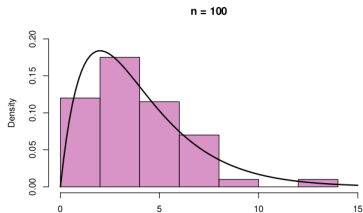
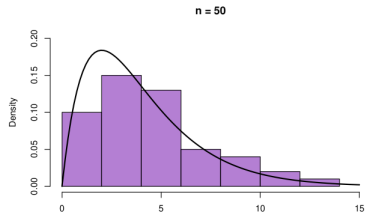
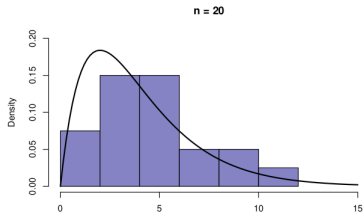
Simulation – Example of Monte Carlo

- Let $X_1, \dots, X_4 \stackrel{iid}{\sim} \text{Exp}(\lambda)$.
- Test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.
- $\bar{X}$ can estimate μ , but what is the **distribution under H_0** ?
 - Definitely not t (data not normal)
 - Not normal (no sufficient data for CLT)

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- › $\bar{X}$ can estimate μ , but what is the distribution under H_0 ?
 - › Definitely not t (data not normal)
 - › Not normal (no sufficient data for CLT)
- › Solution
 - › Simulate $X_1^*, \dots, X_4^* \stackrel{iid}{\sim} \text{Exp}(1/\mu_0)$
 - › Compute $\bar{X}^*$ each time
 - › This gives a large sample of what the distribution of $\bar{X}$ is under H_0 !

Simulation – Illustration



To get n samples from a distribution, in **R** we use pseudo-random number generators

Pseudo-Random Number Generators – Excursus

“Anyone who considers *arithmetical methods* of producing *random digits* is, of course, in a state of sin”

J. von Neumann

“Random numbers should not be generated with a *method chosen at random*”

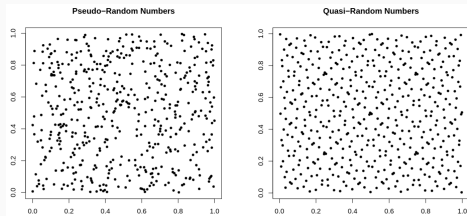
D. Knuth

➤ Pseudo-random numbers are not really random

- They are generated by a *deterministic* algorithm
- Given the same starting point of the algorithm, a generated sequence can be *repeated* (this is key for *reproducibility*; `set.seed()`)

🤖 However, to us humans, they are *indistinguishable* from random ones

- The quality of RNGs can be tested (*Diehard* & *Dieharder* test batteries)
- Compare *pseudo*- and *quasi*-random numbers:



Bootstrap

Bootstrap – Motivation

- Recall: To test an hypothesis H_0 , we require the distribution the test statistic $T_n(X_1, \dots, X_n)$ under H_0 .
- Example: The t-test with $H_0 : \mu = \mu_0$.
 - $T_n(X_1, \dots, X_n) = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}}$.
 - For normal data, under H_0 , $T_n \sim t_{n-1}$.
 - t-approximation is flawed if data is not normal!
- How do we draw conclusions about T_n if we don't have the distribution?
- Core idea: Draw samples of the RV T_n^* with a distribution that approximates the distribution of T_n .

Bootstrap – Idea

- › Bootstrap refers to the idiom to pull oneself up by one's bootstraps
 - › *To begin an enterprise or recover from a setback without any outside help; to succeed only by one's own efforts or abilities*
 - › The term is associated with Baron Münchhausen

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- › Suppose $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P$, where P (potentially) **unknown**
 - › Quantity of interest: $T_n(X_1, \dots, X_n) \sim Q_P (:= Q_P(T_n))$
 - › T_n is e.g. an **estimator** or a **test statistic**
 - › Q_P depends on P and is also (typically) **unknown**
 - › We are interested in **properties** of Q_P (particularly, its variance)

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 - › Q_P depends on P and is also (typically) **unknown**
 - › We are interested in **properties** of Q_P (particularly, its variance)
- › General idea: replace P with an **estimator** $\tilde{P}_n$, and Q_P with $Q_{\tilde{P}_n}$
 - › Note **dependence** on the data via $\tilde{P}_n$
 - › Given a **realized sample** $x_1, \dots, x_n$ from $X_1, \dots, X_n$, estimate $\tilde{P}_n$ is fixed
 - › We can generate a **bootstrap sample** $X_1^*, \dots, X_n^*$ from $\tilde{P}_n$ via **simulation**
 - › A **bootstrap value** of T_n is then given by $T_n^* := T_n(X_1^*, \dots, X_n^*)$

Bootstrap – Real World vs. Bootstrap World

Real World

Unknown
distribution

Realized
sample

$P \longrightarrow x_1, \dots, x_n$

Quantity
of interest

$T_n(x_1, \dots, x_n)$

Bootstrap World

Estimated
distribution

Bootstrap
sample

$\tilde{P}_n \longrightarrow x_1^*, \dots, x_n^*$

Bootstrap
value

$T_n^* := T_n(x_1^*, \dots, x_n^*)$

Bootstrap – Examples

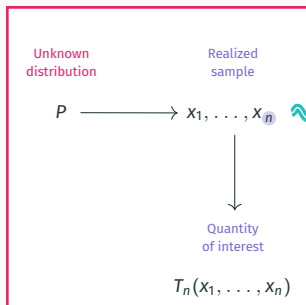
- › Estimator T_n for the theoretical expectation of a distribution
 - › Consider $T_n = \bar{X}$: sample mean of random sample $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P$
 - › P unknown $\rightsquigarrow Q_P$ unknown
- ② Although asymptotically the distribution of the sample mean is what?
 - › Unknown P is estimated via $\tilde{P}_n$, and the aim is to approximate $Q_{\tilde{P}_n}$
 - › Bootstrap: compute $T_n^* = \bar{X}^*$ a large number of times (from a large number B of simulated samples from $\tilde{P}_n$) to approximate $Q_{\tilde{P}_n}$

Bootstrap – Examples

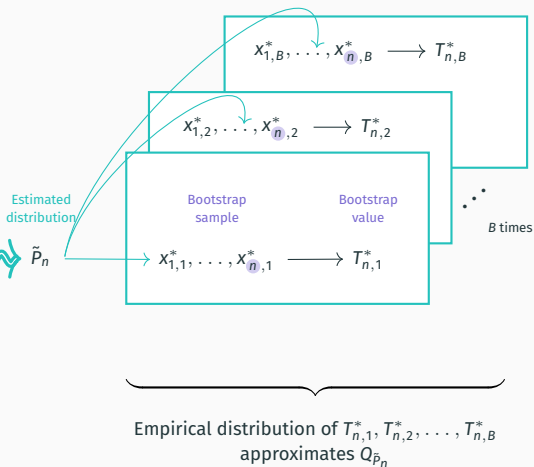
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- › Test statistic $T_n = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ of the one-sample t-test (location test)
 - › If $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P = N(\mu, \sigma^2)$, then $T_n \sim Q_P = t_{n-1}$ under $H_0 : \mu = \mu_0$
 - › $P \neq N$ (and possibly unknown) $\rightsquigarrow$ t approximation flawed
 - › If $P \neq N$, P is estimated via $\tilde{P}_n$, and the aim is to approximate $Q_{\tilde{P}_n}$
 - › Bootstrap: compute $T_n^* = \frac{\bar{X}^* - \mu_0}{S/\sqrt{n}}$ a large number of times (from a large number B of simulated samples from $\tilde{P}_n$) to estimate $Q_{\tilde{P}_n}$

Bootstrap – Real World vs. Bootstrap World

Real World



Bootstrap World



Empirical Bootstrap vs. Parametric Bootstrap

Empirical Bootstrap vs. Parametric Bootstrap

- There are two types of frequently used estimators $\tilde{P}_n$ for **unknown** P
- Use **empirical distribution** $\hat{P}_n \rightsquigarrow$ **empirical** bootstrap
 - This uses **resampling with replacement** from the **realized** sample

original sample:					$\rightsquigarrow T_n$
bootstrap sample 1:					$\rightsquigarrow T_{n,1}^*$
bootstrap sample 2:					$\rightsquigarrow T_{n,2}^*$
bootstrap sample 3:					$\rightsquigarrow T_{n,3}^*$
bootstrap sample 4:					$\rightsquigarrow T_{n,4}^*$
...					...

$\rightsquigarrow$ Histogram of $T_{n,1}^*, T_{n,2}^*, \dots, T_{n,B}^*$ approximates $Q_{\hat{P}_n}$

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- Use **parametric estimator** $P_{\hat{\theta}_n} \rightsquigarrow$ **parametric** bootstrap
 - Appropriate if P is known to belong to a parametric family
 - P is then known **up to** parameter(s) θ
 - First, need estimator $\hat{\theta}_n$ of θ (from the **realized** sample $x_1, \dots, x_n$)

Empirical Bootstrap – Scheme

Aim: Obtain an estimate of $Q_{\hat{P}_n}$, instead of the **unknown** Q_P

Input: **Sample** $x_1, \dots, x_n$ of realizations from $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P$

Step 1: obtain empirical estimator of P

└ Using $x_1, \dots, x_n$, estimate P by $\hat{P}_n$

Step 2: bootstrap

└ **for** $i = 1, \dots, B$ **do**

└└ Generate **bootstrap sample** $X_{1,i}^*, \dots, X_{n,i}^* \stackrel{i.i.d.}{\sim} \hat{P}_n$ // Resample with replacement from realizations

└└ Compute $T_{n,i}^* = T_n(X_{1,i}^*, \dots, X_{n,i}^*)$

Step 3: bootstrap estimate for the distribution of T_n

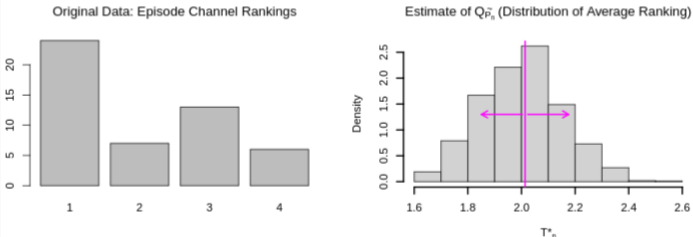
└ Estimate $Q_{\hat{P}_n}$ via the empirical distribution of $T_{n,1}^*, \dots, T_{n,B}^*$

Output: Bootstrap estimate of $Q_{\hat{P}_n}$

Empirical Bootstrap – Illustration

```
### Data
# Package bakeoff: data from 'The Great British Bake Off'
# Distribution of the average episode weekly ranking (across channels)?
rankings <- bakeoff::ratings_raw$channels_rank
rankings <- rankings[complete.cases(rankings)]

### Bootstrap
B <- 10^3
T_star <- numeric(B)
for(i in 1:B){
  # Check ?"sample"
  # The default for argument size is the number of items in the first argument
  X_star <- sample(rankings, replace = TRUE)
  T_star[i] <- mean(X_star)
}
```



```
## [1] "Mean of the empirical distribution of bootstrap values: 2.013"
```

```
## [1] "Standard deviation of the empirical distribution of bootstrap values: 0.152"
```

Parametric Bootstrap – Scheme

Aim: Obtain an estimate of $Q_{P_{\hat{\theta}_n}}$, instead of the **unknown** Q_{P_θ}

Input: **Sample** $x_1, \dots, x_n$ of realizations from $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} P_\theta$

Step 1: obtain parametric estimator of P

└ Using $x_1, \dots, x_n$, estimate θ by $\hat{\theta}_n$ and P_θ by $P_{\hat{\theta}_n}$

Step 2: bootstrap

└ **for** $i = 1, \dots, B$ **do**

└└ Generate **bootstrap sample** $X_{1,i}^*, \dots, X_{n,i}^* \stackrel{i.i.d.}{\sim} P_{\hat{\theta}_n}$ // Sample from estimated parametric distribution

└└ Compute $T_{n,i}^* = T_n(X_{1,i}^*, \dots, X_{n,i}^*)$

Step 3: bootstrap estimate for the distribution of T_n

└ Estimate $Q_{P_{\hat{\theta}_n}}$ via the empirical distribution of $T_{n,1}^*, \dots, T_{n,B}^*$

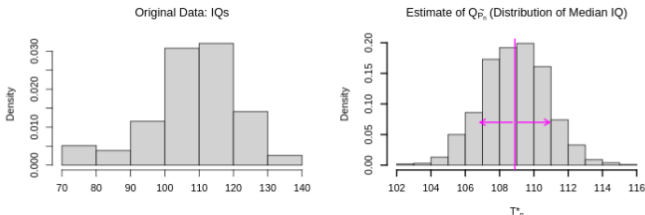
Output: Bootstrap estimate of $Q_{P_{\hat{\theta}_n}}$

Parametric Bootstrap – Illustration

```
### Data
# Package openintro: data and functions from 'OpenIntro' resources
# Distribution of median IQ?
iqs <- openintro::gpa_iq$iq
# any(is.na(iqs)) # FALSE

### Parametric Estimator of P
# IQ is typically considered to be normally distributed
# Need estimates for expectation and standard deviation
mu_hat <- mean(iqs)
sigma_hat <- sd(iqs)

### Bootstrap
B <- 10^3
T_star <- numeric(B)
n <- length(iqs)
for(i in 1:B){
  X_star <- rnorm(n, mean = mu_hat, sd = sigma_hat)
  T_star[i] <- median(X_star)
}
```



```
## [1] "Mean of the empirical distribution of bootstrap values: 108.891"
```

```
## [1] "Standard deviation of the empirical distribution of bootstrap values: 1.878"
```

Bootstrap Errors

- › Error source 1: estimation of P via $\tilde{P}_n$ (and of Q_P via $Q_{\tilde{P}_n}$)
 - › This error is (in some sense) **unavoidable**
 - › Its magnitude depends on the specific T_n and the chosen estimator $\tilde{P}_n$
 - › **Wrong** parametric distribution $P_\theta \rightsquigarrow$ **Big** error
 - › Using the empirical distribution $\hat{P}_n$ is (usually) a **safer** choice

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 - › Using the empirical distribution $\hat{P}_n$ is (usually) a **safer** choice
- › Error source 2: estimation of $Q_{\tilde{P}_n}$ via the empirical distribution of

$$T_{n,1}^*, \dots, T_{n,B}^* \stackrel{i.i.d.}{\sim} Q_{\tilde{P}_n}$$

- › This error can be **controlled**
- › Its magnitude depends on B (number of bootstrap samples)
- › **Large** $B \rightsquigarrow$ **Small** error
- › In theory, this error can be made **arbitrarily small** by increasing B
- › In practice, one has to consider the available **computing power** and the necessary **computational runtime**
- › The **severity** of this practical limitation, in turn, depends on the **complexity** of T_n

Summary and wrap-up

Summary

- › True randomness **does not exist** in simulations, we use **pseudo-random numbers** or **quasi-random numbers** instead.
- › In a Bootstrap sampling method, we
 - › Sample $X_1^*, \dots, X_n^*$ from an approximation $\tilde{P}$ of the underlying distribution.
 - › Obtain samples of a statistic $T_n(X_1^*, \dots, X_n^*)$ to draw conclusions about $T(X_1, \dots, X_n)$.
 - › The **empirical Bootstrap** uses the empirical distribution as $\tilde{P}$.
 - › The **parametric Bootstrap** uses a parametric estimate for $\tilde{P}$.

Wrap-up

- › What now?
 - › Continue working on assignment 3
 - › Deadline: 10 March 23:59
 - › All material now covered
- › Next week
 - › Bootstrap confidence intervals
 - › Bootstrap tests
 - › Limitations of the bootstrap